14D
CHEMICAL KINETICS

UNIT 02: HOW REACTIONS HAPPEN

CHEMISTRY 1310

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LEARNING OBJECTIVES

CHAPTER 14

1. Express reactions rates in relation to reaction stoichiometry.
2. Use experimental data to write rate laws for chemical reactions.
3. Predict the concentration and time values of reactions using first- and second-order integrated rate laws.
4. Graph data to identify the order of a reaction.
5. Define and apply the terms half-life, first-order reaction, and second-order reaction.
6. Explain collision theory using appropriate terminology.
7. Apply the Arrhenius equation to calculate activation energy and/or the value of the rate constant at various temperatures.
8. Define and discuss reaction mechanisms.
9. Write rate laws for elementary steps.
10. Identify intermediates in a reaction mechanism.
11. Discuss how catalysts work to speed up chemical reactions

Reaction Rates and Temperature: Activation Energy

14.4

Rates of chemical reactions depend on temperature.

Qualitatively, we can say that most reactions go *faster* as temperature *increases*.

Chemical reactions involve the rearrangement of atoms in the reactants and products.

- *Some energy input* is always required to start reactions.
- **Activation energy (E_a)**: The threshold/minimum amount of energy that must first be overcome in order for a reaction to occur.
  - The magnitude of the E_a is *inversely* proportional to the magnitude of k.
  - As E_a increases, k decreases, and the reaction rate slows down.

Reaction Rates and Temperature: Activation Energy

14.4

At the moment of an *effective collision*, atoms will begin to rearrange.

- For an instant, a high-energy and unstable species called the **transition state** or **activated complex** exists.
- The difference in energy between the reactants and the transition state is E_a.
- E_a represents a *barrier* to product formation.

Reaction Rates and Temperature: Activation Energy

14.4

Collision theory: A model for describing molecular motion that helps us explain rates of chemical reactions. *Principles of collision theory:*

1) Molecules must physically collide to react.
2) Collisions only result in reactions *if* they occur with:
   - A molecular orientation that facilitates formation of desired products
   - Sufficient kinetic energy to overcome the activation energy (E_a)

The Arrhenius equation describes the relationship among molecular orientation, kinetic energy, and the rate (constant) of a reaction.

Reaction Rates and Temperature: Activation Energy

14.4

Frequency factor (A)

- How *often* collisions occur. The rate of collisions can be increased by increasing the concentration and, to a smaller extent, by increasing the temperature.
- The probability that collision occur with the reactants in the *appropriate* orientation.

Only a certain percentage of collisions will have the correct orientation for a reaction occur.

For example, consider the *three* orientations for the reaction:

$$2\,NO \rightarrow N_2 + O_2$$

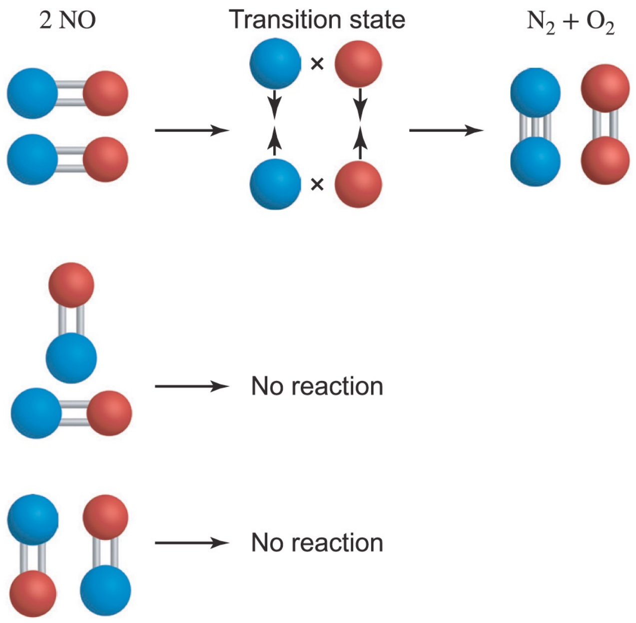

FIGURE 14.17 Only the top collision is "effective" and leads to product formation.

Reaction Rates and Temperature: Activation Energy

14.4

Temperature (T)

- Average kinetic energy of molecules is proportional to the absolute temperature.
- When E_a *is large*, the percentage of molecules that have or exceed this amount of energy is low.
- As T *increases*, the percentage of molecules that have or exceed E_a *increases*.

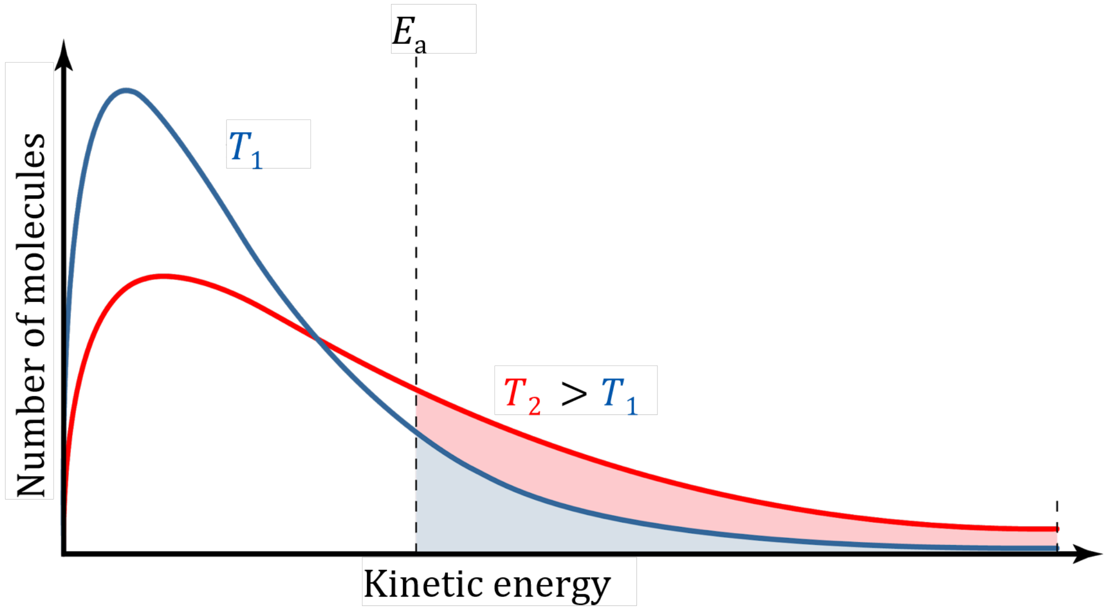

FIGURE 14.18

Reaction Rates and Temperature: Activation Energy

14.4

There are different forms of the Arrhenius equation.

IN-CLASS QUESTION 1

Learning Objective(s): 14.7

Calculate the activation energy (in kJ/mol) for a reaction if the rate constant for the reaction *doubles* when the temperature increases from 295 K to 305 K.

Reaction Mechanisms

14.5

A **reaction mechanism** is the sequence of single molecular steps that account for the process by which reactants are converted to products during a chemical reaction.

The single molecular steps are called **elementary steps** (or elementary reactions).

- Elementary steps in a mechanism are summed to give the overall reaction.
- Each elementary step has its own *elementary rate law*.
- The rate of the *overall* reaction is limited by the rate of the *slowest elementary step* in the mechanism.
  - This slowest step is called the **rate-limiting step** or **rate-determining step**.
  - The rate-limiting step is the one with the greatest activation energy (E_a).
  - The elementary rate law for the slowest step informs the overall rate law.

Reaction Mechanisms

14.5

Consider the **overall reaction:** AB + C → AC + B

Step 1: Reactant AB separates into A and B.

AB → A + B

Step 2: A combines with C to form the product AC.

A + C → AC

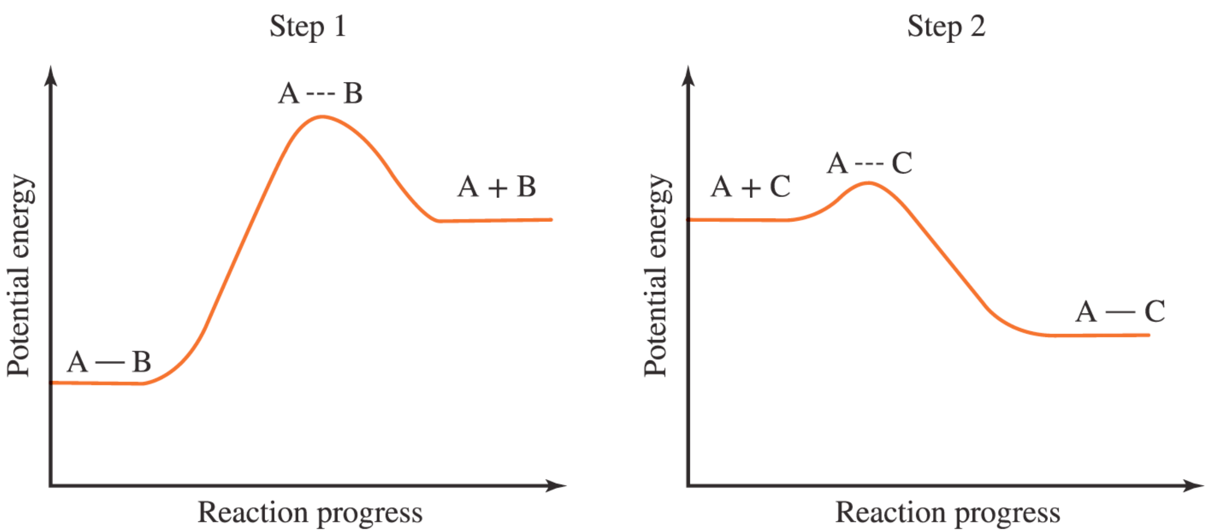

FIGURE 14.19

Each elementary step represents an individual molecular event, making it possible to *write the elementary rate laws directly from the balanced equation for the elementary step*.

Each elementary step has its own transition state and its own activation energy.

Reaction Mechanisms

14.5

Intermediates are species that are produced in an early elementary step and subsequently consumed in a later elementary step.

- Intermediates do ***NOT*** appear in the overall chemical equation.

In the previous example, A is an intermediate.

- A is *produced* in elementary step 1.
- A is *consumed* in elementary step 2.
- A does ***not*** appear in the overall reaction.

$$AB + C \rightarrow AC + B$$

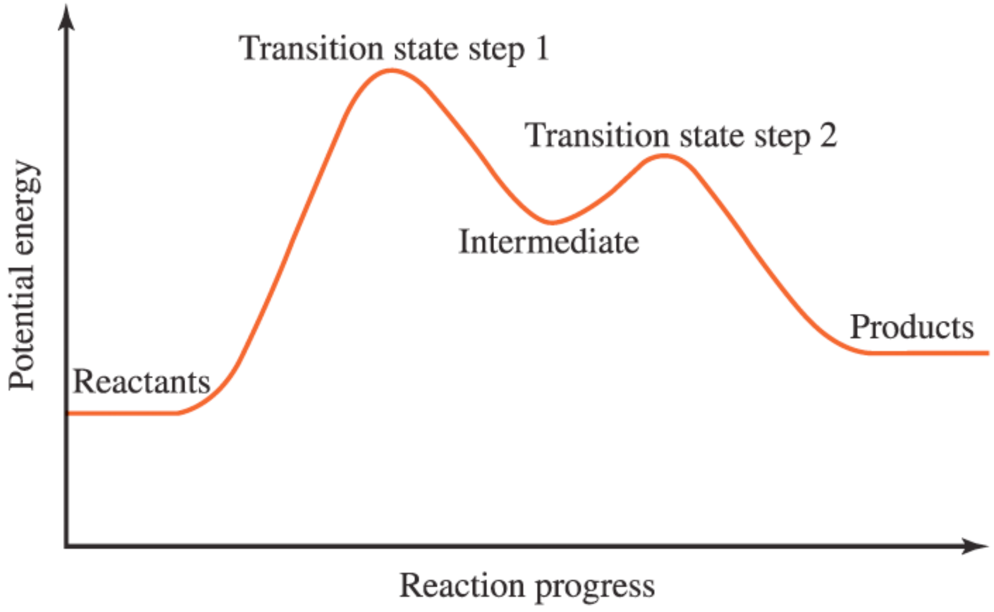

FIGURE 14.20

EXAMPLE PROBLEM

Learning Objective(s): 14.9

Write the overall reaction given the following mechanism. Then, identify the reactants, products, and any intermediates if they exist.

Step 1: $NO_2(g) + NO_2(g) \xrightarrow{k_1} NO(g) + NO_3(g)$ *(slow)*

Step 2: $NO_3(g) + CO(g) \xrightarrow{k_2} NO_2(g) + CO_2(g)$ *(fast)*

Overall:

IN-CLASS QUESTION 2

Learning Objective(s): 14.9

Given the mechanism, what is the expected overall (observed) rate law?

- ❑ Rate = $k[NO_2][CO]$
- ❑ Rate = $k[NO_2]^2[CO]$
- ❑ Rate = $k[NO_2]^2$
- ❑ Rate = $k[NO_3][CO]$
- ❑ Rate = $k[NO_2]^2[NO_3][CO]$

Step 1: $NO_2(g) + NO_2(g) \xrightarrow{k_1} NO(g) + NO_3(g)$ *(slow)*

Step 2: $NO_3(g) + CO(g) \xrightarrow{k_2} NO_2(g) + CO_2(g)$ *(fast)*

Catalysis

14.6

Catalysts provide an *alternative* pathway (mechanism) for the reaction to occur.

- The alternative pathway follows a *different* set of elementary steps.
- The catalyzed pathway has a lower E_a, which speeds up the reaction.
- Catalysts are ***not*** consumed by reactions. Rather, they are reactants in an early elementary step and are re-generated in a subsequent elementary step.

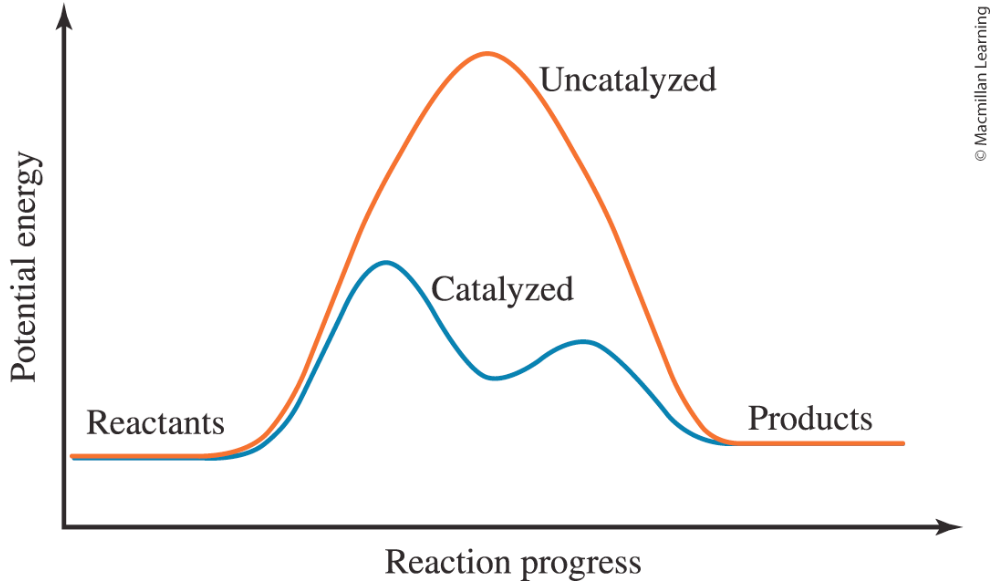

FIGURE 14.23

IN-CLASS QUESTION 3

Learning Objective(s): 14.10, 14.11

Write the overall reaction and identify any intermediates and/or catalysts.

Step 1: $H_2O_2(aq) + 2\ HBr(aq) \rightarrow Br_2(aq) + 2\ H_2O(\ell)$

Step 2: $Br_2(aq) + H_2O_2(aq) \rightarrow 2\ HBr(aq) + O_2(g)$

Overall:

BONUS QUESTION

Assessed on: Fall 2021, Exam 3

For the decomposition reaction below, the rate constant (k) was determined at several different temperatures. A plot of $\ln(k)$ vs. $1/T$ was generated and the data was fitted to a linear equation.

$$2\ N_2O_5(g) \rightarrow 2\ N_2O_4(g) + O_2(g)$$

Estimate the activation energy (in kJ/mol) for the reaction.